Star Selection Games and the Pixley-Roy Hyperspace

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Introduction

Selection Principles

Definition

$\mathsf{S}_1(\mathcal{A},\mathcal{B}) := \forall A \in \mathcal{A}^\omega\ \exists x \in \prod_{n\in\omega} A_n\ (\{x_n : n \in \omega\} \in \mathcal{B})$

Definition

$\mathsf{S}_{\mathrm{fin}}(\mathcal{A},\mathcal{B}) := \forall A \in \mathcal{A}^\omega\ \exists F \in \prod_{n\in\omega} [A_n]^{<\omega}\ (\bigcup_{n\in\omega} F_n \in \mathcal{B})$

At times, it will be convenient to use formulations in terms of sequences:

Definition

$\vec{\mathsf{S}}_1(\mathcal{A}, \mathcal{B}) := \forall A \in \mathcal{A}^\omega \ \exists B \in \prod_{n\in\omega} A_n \ (B \in \mathcal{B})$

For finite selections, there is a bit more liberty, but we tentatively settle with the following:

Definition

$\vec{\mathsf{S}}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B}) := \forall A \in \mathcal{A}^\omega \ \exists F \in \prod_{n\in\omega} [A_n]^{<\omega} \ (\langle \bigcup F_n : n \in \omega \rangle \in \mathcal{B})$

These selection principles give rise to selection games:

Definition

$\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ is a game of two players, One and Two.

- In the nth inning, One plays $A_n \in \mathcal{A}$ and Two responds with $x_n \in A_n$.
- If $\{x_n : n \in \omega\} \in \mathcal{B}$, then Two wins. Otherwise, One wins.

The finite selection game $\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$ and the games $\vec{\mathsf{G}}_1(\mathcal{A}, \mathcal{B})$ and $\vec{\mathsf{G}}_{\text{fin}}(\mathcal{A}, \mathcal{B})$ are analogously defined.

Full-information Strategies

- A *winning strategy for One* in $\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ is a function $\sigma : (\bigcup \mathcal{A})^{<\omega} \to \mathcal{A}$ so that, whenever $x_n \in \sigma\langle x_m : m < n\rangle$ for all $n \in \omega$, $\{x_n : n \in \omega\} \notin \mathcal{B}$.
  - If One has a winning strategy, we write $\mathrm{I} \uparrow \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.
- A *winning strategy for Two* in $\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ is a function $\tau : \mathcal{A}^{<\omega} \to \bigcup \mathcal{A}$ so that, whenever $A_n \in \mathcal{A}$ for all $n \in \omega$, $\{\tau\langle A_m : m < n\rangle : n \in \omega\} \in \mathcal{B}$.
  - If Two has a winning strategy, we write $\mathrm{II} \uparrow \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.

Limited-information Strategies

- A *predetermined strategy* for One is a strategy which only considers the current turn number. Formally it is a function $\sigma : \omega \to \mathcal{A}$.
  - If One has a winning predetermined strategy, we write $\mathrm{I} \underset{\text{pre}}{\uparrow} \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.
- A *Markov strategy* for Two is a strategy which only considers the most recent move of player One and the current turn number. Formally it is a function $\tau : \mathcal{A} \times \omega \to \bigcup \mathcal{A}$.
  - If Two has a winning Markov strategy, we write $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.

All of these strategy types can be defined analogously for the other selection games we've introduced.

Transfer of Strategic Information

Definition

For two selection games $\mathcal{G}$ and $\mathcal{H}$, we define $\mathcal{G} \leqslant_{\mathrm{II}} \mathcal{H}$ to be the conjunction

- $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G} \implies \mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{H}$
- $\mathrm{II} \uparrow \mathcal{G} \implies \mathrm{II} \uparrow \mathcal{H}$
- $\mathrm{I} \not\uparrow \mathcal{G} \implies \mathrm{I} \not\uparrow \mathcal{H}$
- $\mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{G} \implies \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{H}$

This $\leqslant_{\mathrm{II}}$ is transitive and two selection games $\mathcal{G}$ and $\mathcal{H}$ are equivalent if and only if $\mathcal{G} \leqslant_{\mathrm{II}} \mathcal{H}$ and $\mathcal{H} \leqslant_{\mathrm{II}} \mathcal{G}$.

Observe that

$$\begin{aligned} \mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\mathcal{A},\mathcal{B}) &\implies \mathrm{II} \uparrow \mathsf{G}_1(\mathcal{A},\mathcal{B}) \\ &\implies \mathrm{I} \not\uparrow \mathsf{G}_1(\mathcal{A},\mathcal{B}) \\ &\implies \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathsf{G}_1(\mathcal{A},\mathcal{B}) \end{aligned}$$

and that $\mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathsf{G}_1(\mathcal{A},\mathcal{B})$ implies a Lindelöf-like principle: for any $A \in \mathcal{A}$, there is $B \in [\mathcal{A}]^{\leqslant\omega}$ so that $B \in \mathcal{B}$.

Theorem (C. and Holshouser, 2019)

Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{D}$ be collections. Suppose there are functions

- $\overleftarrow{T}_{\mathrm{I},n} : \mathcal{B} \to \mathcal{A}$ and
- $\overrightarrow{T}_{\mathrm{II},n} : \bigcup \mathcal{A} \times \mathcal{B} \to \bigcup \mathcal{B}$

for each $n \in \omega$ so that

1. $x \in \overleftarrow{T}_{\mathrm{I},n}(B) \implies \overrightarrow{T}_{\mathrm{II},n}(x, B) \in B$ and
2. $x \in \prod_{n\in\omega} \overleftarrow{T}_{\mathrm{I},n}(B_n) \wedge \{x_n : n \in \omega\} \in \mathcal{C}$
$$\implies \left\{\overrightarrow{T}_{\mathrm{II},n}(x_n, B_n) : n \in \omega\right\} \in \mathcal{D}.$$

Then $\mathsf{G}_1(\mathcal{A}, \mathcal{C}) \leqslant_{\mathrm{II}} \mathsf{G}_1(\mathcal{B}, \mathcal{D})$.

There is an analogous result for games of finite selection.

Stars and Star Selections

Definition

For a set X, a collection $\mathcal{R}$ of subsets of X, and $A \subseteq X$,

$$\mathrm{St}(A, \mathcal{R}) := \bigcup \{B \in \mathcal{R} : A \cap B \neq \emptyset\}.$$

We let $\mathcal{O}_X$ denote the set of open covers of a space X; when the space is understood from context, we use, simply, $\mathcal{O}$.

Star Selections (Kočinac, 1999)

- $\mathsf{S}_1^*(\mathcal{O}, \mathcal{B})$: for each sequence of open covers $\mathscr{U}_n$, there are open sets $U_n \in \mathscr{U}_n$ so that $\{\mathrm{St}(U_n, \mathscr{U}_n) : n \in \omega\} \in \mathcal{B}$.
- $\mathsf{SS}_{\mathcal{A}}^*(\mathcal{O}, \mathcal{B})$: for each sequence of open covers $\mathscr{U}_n$, there is a sequence of $A_n \in \mathcal{A}$ so that $\{\mathrm{St}(A_n, \mathscr{U}_n) : n \in \omega\} \in \mathcal{B}$.

Star covering properties appeared at least as early as 1991 (E.K. van Douwen, G.M. Reed, A.W. Roscoe and I.J. Tree).

Constellations and Galaxies

- If $\mathscr{U}$ is an open cover of X and $\mathcal{A}$ is a collection of subsets of X, then $\mathrm{Cons}(\mathcal{A}, \mathscr{U}) = \{\mathrm{St}(A, \mathscr{U}) : A \in \mathcal{A}\}$.
- If $\mathscr{C}$ is a collection of open covers of X and $f : \mathscr{C} \to \mathcal{P}(\mathcal{P}(X))$, then $\mathrm{Gal}(f, \mathscr{C}) = \{\mathrm{Cons}(f(\mathscr{U}), \mathscr{U}) : \mathscr{U} \in \mathscr{C}\}$.

Note that

- $\mathsf{S}_1^*(\mathcal{O}, \mathcal{B})$ is equivalent to $\mathsf{S}_1(\mathrm{Gal}(\mathrm{id}, \mathcal{O}), \mathcal{B})$ and that
- $\mathsf{SS}_{\mathcal{A}}^*(\mathcal{O}, \mathcal{B})$ is equivalent to $\mathsf{S}_1(\mathrm{Gal}(\mathcal{A}, \mathcal{O}), \mathcal{B})$.

With this equivalence in mind, we will apply our translation results to $\mathsf{G}_1^*(\mathcal{O}, \mathcal{B})$.

The relationship we will establish involves the Pixley-Roy hyperspace and ω-covers.

Definition (Pixley and Roy, 1969)

For a topological space X, the *Pixley-Roy hyperspace* $\mathrm{PR}(X)$ is the the set $[X]^{<\omega}$ endowed with the topology generated by

$$[F, U] = \{E \in [X]^{<\omega} : F \subseteq E \subseteq U\}$$

where $F \in [X]^{\omega}$ and $U \subseteq X$ is open with $F \subseteq U$.

Definition

A non-trivial open cover $\mathscr{U}$ is an *ω-cover* if, for every $F \in [X]^{<\omega}$, there is some $U \in \mathscr{U}$ so that $F \subseteq U$.
For a space X, let Ω_X denote the collection of ω-covers.

The main theorem we will establish here is

Theorem (C. and Holshouser, 2022)

Assume X is regular. Then, for $\square \in \{1, \mathrm{fin}\}$,

$$\mathsf{G}^*_\square(\mathcal{O}_{\mathrm{PR}(X)}, \mathcal{O}_{\mathrm{PR}(X)}) \leqslant_{\mathrm{II}} \mathsf{G}_\square(\Omega_X, \Omega_X)$$

and, in general, the inequality does not reverse.

As a corollary,

Corollary (Sakai, 2014)

Suppose X is regular. Then

$$\mathsf{S}^*_{\mathrm{fin}}(\mathcal{O}_{\mathrm{PR}(X)}, \mathcal{O}_{\mathrm{PR}(X)}) \implies \mathsf{S}_{\mathrm{fin}}(\Omega_X, \Omega_X).$$

In Sakai's original language, if $\mathrm{PR}(X)$ is star-Menger, then every finite power of X is Menger.

We first sketch some necessary preliminary results.

Definition

The *star-Lindelöf degree* $L_{\mathrm{st}}(X)$ of a space X is the least cardinal κ so that, for any $\mathscr{U} \in \mathcal{O}_X$, there exists $\mathscr{V} \in [\mathscr{U}]^{\leqslant\kappa}$ so that

$$\{\mathrm{St}(V, \mathscr{U}) : V \in \mathscr{V}\} \in \mathcal{O}_X.$$

Definition

The *discrete cellularity* $dc(X)$ of a space X is the supremum over all cardinals κ so that there exists a discrete κ-sized family of open subsets of X.

Lemma

For any space X with a basis consisting of clopen sets, $dc(X) \leqslant L_{\mathrm{st}}(X)$.

proof sketch

- Let $\{U_\alpha : \alpha < \kappa\}$ be a discrete family of open sets.
- Without loss of generality, assume that each U_α is clopen.
- By discreteness, $W := X \setminus \bigcup\{U_\alpha : \alpha < \kappa\}$ is open, so $\mathscr{U} = \{U_\alpha : \alpha < \kappa\} \cup \{W\}$ is an open cover of X consisting of pair-wise disjoint sets.
- Hence, for any $V \in \mathscr{U}$, $\mathrm{St}(V, \mathscr{U}) = V$.
- Clearly, no strict sub-collection $\mathscr{V}$ of $\mathscr{U}$ has the property that $\{\mathrm{St}(V, \mathscr{U}) : V \in \mathscr{V}\} \in \mathcal{O}_X$. $\square$

Consequently, for any space X,

$$dc(\mathrm{PR}(X)) \leqslant L_{\mathrm{st}}(\mathrm{PR}(X)).$$

Definition

The *ω-Lindelöf degree* $L_\omega(X)$ of a space X is the minimal cardinal κ so that any ω-cover admits a $\leqslant \kappa$-sized sub-ω-cover.

- $hL(X)$ denotes the *hereditary Lindelöf degree* of X, the supremum of the Lindelöf degrees of subspaces (equivalently, open subspaces) of X.
- $hL_\omega(X)$ denotes the *hereditary ω-Lindelöf degree* of X, the supremum of the ω-Lindelöf degrees of open subspaces of X.

By taking an open cover $\mathscr{U}$ of X and considering the auxiliary open cover of finite unions of the elements of $\mathscr{U}$, we can see that $L(X) \leqslant L_\omega(X)$; hence, $hL(X) \leqslant hL_\omega(X)$.

Lemma

For a space X, $hL_\omega(X) \leqslant dc(\mathrm{PR}(X))$.

proof sketch

- Show that $L_\omega(W) \leqslant dc(\mathrm{PR}(X))$ for any open $W \subseteq X$.
- Note that $\mathrm{PR}(W)$ is clopen in $\mathrm{PR}(X)$, so $dc(\mathrm{PR}(W)) \leqslant dc(\mathrm{PR}(X))$.
- Let $\mathscr{U} = \{U_\alpha : \alpha < \lambda\}$ be a non-trivial ω-cover of W.
- Define
$$V_\alpha = \mathrm{PR}(U_\alpha) \setminus \bigcup_{\beta<\alpha} V_\beta.$$
- The family $\{V_\alpha : \alpha < \lambda\}$ is a discrete family of open sets, so we can use the discrete cellularity to produce an ω-cover of W of the desired size. $\square$

Definition

Let $\psi_k(X)$ denote the minimal cardinal κ so that every compact subset of X can be written as the $\leqslant \kappa$-sized intersection of open sets.

Lemma

For a regular space X, $\psi_k(X) \leqslant hL(X)$. Consequently,

$$\psi_k(X) \leqslant L_{\mathrm{st}}(\mathrm{PR}(X)).$$

We now sketch a proof, in the single selection case for notational simplicity, of the theorem.

- We may assume that $\mathrm{PR}(X)$ is star-Lindelöf; hence, every finite set F can be written as $F = \bigcap_{n\in\omega} W_{F,n}$ where $\{W_{F,n} : n \in \omega\}$ is a descending family of open sets.
- Given an ω-cover $\mathscr{U}$ and $F \in [X]^{<\omega}$, let $\gamma(F, \mathscr{U}) \in \mathscr{U}$ be so that $F \subseteq \gamma(F, \mathscr{U})$. Then define

$$\mathbf{W}(F, \mathscr{U}, n) = [F, \gamma(F, \mathscr{U}) \cap W_{F,n}].$$

- Define, for $\mathscr{U} \in \Omega_X$ and $n \in \omega$,

$$\mathscr{V}_{\mathscr{U},n} = \{\mathbf{W}(F, \mathscr{U}, n) : F \in \mathrm{PR}(X)\},$$

  an open cover $\mathrm{PR}(X)$.

- Define $\overleftarrow{T}_{\mathrm{I},n} : \Omega_X \to \mathrm{Gal}(\mathrm{id}, \mathcal{O}_{\mathrm{PR}(X)})$ by

$$\overleftarrow{T}_{\mathrm{I},n}(\mathscr{U}) = \mathrm{Cons}(\mathscr{V}_{\mathscr{U},n}, \mathscr{V}_{\mathscr{U},n}).$$

- For $\mathbb{W} \in \mathrm{Cons}(\mathscr{V}_{\mathscr{U},n}, \mathscr{V}_{\mathscr{U},n})$, let $\chi_{\mathscr{U},n}(\mathbb{W}) \in \mathrm{PR}(X)$ be so that

$$\mathbb{W} = \mathrm{St}(\mathbf{W}(\chi_{\mathscr{U},n}(\mathbb{W}), \mathscr{U}, n), \mathscr{V}_{\mathscr{U},n}).$$

- As long as $\mathbb{W} \in \mathrm{Cons}(\mathscr{V}_{\mathscr{U},n}, \mathscr{V}_{\mathscr{U},n})$ for $\mathscr{U} \in \Omega_X$, define

$$\overrightarrow{T}_{\mathrm{II},n}(\mathbb{W}, \mathscr{U}) = \gamma(\chi_{\mathscr{U},n}(\mathbb{W}), \mathscr{U}).$$

- Now, to show the translation is successful, suppose we have a sequence $\langle \mathscr{U}_n : n \in \omega \rangle$ of ω-covers of X.
- By using a bijection $\omega^2 \to \omega$, we can assume that Two plays $\langle \mathbb{W}_n : n \in \omega \rangle$, where $\mathbb{W}_n \in \overleftarrow{T}_{\mathrm{I},n}(\mathscr{U}_n)$ for each $n \in \omega$, in such a way that, for every $F \in \mathrm{PR}(X)$ and every $m \in \omega$, there exists $n \geqslant m$ so that $F \in \mathbb{W}_n$.
- To show that $\left\{ \overrightarrow{T}_{\mathrm{II},n}(\mathbb{W}_n, \mathscr{U}_n) : n \in \omega \right\} \in \Omega_X$, let $F \in [X]^{<\omega}$.
- By reducing to an adequate subsequence, we can assume that

$$F \in \mathbb{W}_n = \mathrm{St}(\mathbf{W}(\chi_{\mathscr{U}_n,n}(\mathbb{W}_n), \mathscr{U}_n, n), \mathscr{V}_{\mathscr{U}_n,n})$$

for each n.

- Now, for $n \in \omega$, let $E_n \in \mathrm{PR}(X)$ be so that $F \in \mathbf{W}(E_n, \mathscr{U}_n, n)$ and

$$\mathbf{W}(E_n, \mathscr{U}_n, n) \cap \mathbf{W}(F_n, \mathscr{U}_n, n) \neq \emptyset,$$

  where $F_n := \chi_{\mathscr{U}_n, n}(\mathbb{W}_n)$.
- Since $E_n \subseteq F$ for all $n \in \omega$, by reducing to an adequate subsequence, we can assume E_n is a constant sequence; so let $E = E_n$.
- Now, as $F \subseteq \gamma(E, \mathscr{U}_n) \cap W_{E,n}$ for each $n \in \omega$ and the $W_{E,n}$ are descending, $F = E$.

- Then, let $H \in \mathbf{W}(E, \mathscr{U}_0, 0) \cap \mathbf{W}(F_0, \mathscr{U}_0, 0)$ and behold that

$$\begin{aligned} F = E \subseteq H &\subseteq \gamma(F_0, \mathscr{U}_0) \cap W_{F_0,0} \\ &\subseteq \gamma(F_0, \mathscr{U}_0) \\ &= \overrightarrow{T}_{\mathrm{II},0}(\mathbb{W}_0, \mathscr{U}_0). \end{aligned}$$

The following example demonstrates that

$$\mathsf{G}_{\square}(\Omega_X, \Omega_X) \not\leq_{\mathrm{II}} \mathsf{G}^*_{\square}(\mathcal{O}_{\mathrm{PR}(X)}, \mathcal{O}_{\mathrm{PR}(X)}),$$

in general and, thus, that these games are not equivalent.

Example

Let X be the one-point Lindelöfication of ω_1 with the discrete topology. Then

- $\mathrm{II} \uparrow \mathsf{G}_1(\Omega_X, \Omega_X)$ and
- $\mathrm{PR}(X)$ is not star-Lindelöf which implies that $\mathrm{II} \not\uparrow \mathsf{G}^*_1(\mathcal{O}_{\mathrm{PR}(X)}, \mathcal{O}_{\mathrm{PR}(X)})$.

II $\uparrow \mathsf{G}_1(\Omega_X, \Omega_X)$

- For $n \in \omega$, suppose we have ω-covers $\langle \mathscr{U}_k : k < n \rangle$ and $\langle U_k : k < n \rangle$ defined with $X \setminus U_k = \{x_{k,j} : j \in \omega\}$ and $U_k \in \mathscr{U}_k$ for each $k < n$. Then, for an ω-cover $\mathscr{U}_n$, let Two choose $U_n \in \mathscr{U}_n$ so that $\{\omega_1\} \cup \{x_{k,j} : k < n, j < n\} \subseteq U_n$. Then enumerate $X \setminus U_n = \{x_{n,j} : j \in \omega\}$.
- Check that $\{U_n : n \in \omega\}$ is an ω-cover of X.

$\mathrm{PR}(X)$ is not star-Lindelöf

This follows immediately from the fact that

$$\{[\{\alpha\}, \{\alpha\}] : \alpha \in \omega_1\}$$

is a discrete family of open sets; so

$$\omega_1 \leqslant dc(\mathrm{PR}(X)) \leqslant L_{\mathrm{st}}(\mathrm{PR}(X)).$$

Thank You!